

Enterprise Java – Assignment 1

21-01-2026

1. ATM Transaction Analysis

Write a Java program that accepts a transaction message string (containing letters, digits, spaces, and symbols) and counts the number of letters, digits, spaces, and special characters present in the message.

Sample Input:

Transaction ID: TXN456 Amount ₹2500

Sample Output:

Letters : 16

Digits : 7

Spaces : 4

Special Characters : 2

```
import java.util.Scanner;
```

```
public class ATMTransactionAnalysis {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        // Sample input (you can also take user input)  
        String message = "Transaction ID: TXN456 Amount ₹2500";  
  
        int letters = 0;  
        int digits = 0;  
        int spaces = 0;  
        int special = 0;  
  
        for (int i = 0; i < message.length(); i++) {  
            char ch = message.charAt(i);  
  
            if (Character.isLetter(ch)) {  
                letters++;  
            } else if (Character.isDigit(ch)) {  
                digits++;  
            } else if (Character.isWhitespace(ch)) {  
                spaces++;  
            } else {  
                special++;  
            }  
        }  
    }  
}
```

```

        special++;
    }
}

System.out.println("Letters : " + letters);
System.out.println("Digits : " + digits);
System.out.println("Spaces : " + spaces);
System.out.println("Special Characters : " + special);

sc.close();
}
}

```

2. Mobile Recharge Menu Program

Write a menu-driven Java program to perform the following operations on a mobile recharge plan name:

- i. Replace all characters with * if the length is less than 5
- ii. Concatenate two plan names
- iii. Extract a sub-plan name using substring
- iv. Reverse the plan name

Sample Input:

Enter Plan Name: Data99

Enter choice: 4

Sample Output:

Reversed Plan Name: 99ataD

```

import java.util.Scanner;

public class MobileRechargeMenu {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Plan Name: ");
        String planName = sc.nextLine();

        System.out.println("\nMenu");
        System.out.println("1. Replace all characters with * if length < 5");
        System.out.println("2. Concatenate two plan names");
        System.out.println("3. Extract sub-plan name (substring)");
        System.out.println("4. Reverse the plan name");

        System.out.print("Enter choice: ");
    }
}

```

```
int choice = sc.nextInt();
sc.nextLine(); // consume newline

switch (choice) {

    case 1:
        if (planName.length() < 5) {
            String replaced = "";
            for (int i = 0; i < planName.length(); i++) {
                replaced += "*";
            }
            System.out.println("Modified Plan Name: " + replaced);
        } else {
            System.out.println("Plan name length is 5 or more. No change.");
        }
        break;

    case 2:
        System.out.print("Enter another Plan Name: ");
        String plan2 = sc.nextLine();
        String concatenated = planName.concat(plan2);
        System.out.println("Concatenated Plan Name: " + concatenated);
        break;

    case 3:
        System.out.print("Enter start index: ");
        int start = sc.nextInt();
        System.out.print("Enter end index: ");
        int end = sc.nextInt();

        if (start >= 0 && end <= planName.length() && start < end) {
            String sub = planName.substring(start, end);
            System.out.println("Sub-plan Name: " + sub);
        } else {
            System.out.println("Invalid indices!");
        }
        break;

    case 4:
        String reversed = "";
        for (int i = planName.length() - 1; i >= 0; i--) {
            reversed += planName.charAt(i);
        }
        System.out.println("Reversed Plan Name: " + reversed);
}
```

```

        break;

    default:
        System.out.println("Invalid choice!");
    }

    sc.close();
}
}

```

3. Bus Seat Rearrangement

Write a Java program to swap the first and last seat numbers in an integer array representing occupied bus seats and store the result in a new array.

Sample Input:

12 15 18 21

Sample Output:

21 15 18 12

```

import java.util.Scanner;

public class BusSeatRearrangement {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.println("Enter number of seats:");
        int n = sc.nextInt();

        int[] seats = new int[n];
        int[] newSeats = new int[n];

        System.out.println("Enter seat numbers:");
        for (int i = 0; i < n; i++) {
            seats[i] = sc.nextInt();
        }

        // Copy elements to new array
        for (int i = 0; i < n; i++) {
            newSeats[i] = seats[i];
        }

        // Swap first and last in new array
    }
}

```

```

        if (n > 1) {
            int temp = newSeats[0];
            newSeats[0] = newSeats[n - 1];
            newSeats[n - 1] = temp;
        }

        System.out.println("After Swapping:");
        for (int i = 0; i < n; i++) {
            System.out.print(newSeats[i] + " ");
        }

        sc.close();
    }
}

```

4. Invoice Digit Processing

Write a Java program that reads an invoice number, calculates the sum of its digits, and prints each digit of the sum in words (e.g., 123 → One Two Three).

Sample Input:

458

Sample Output:

Sum of digits: 17

Digits in words: One Seven

```

import java.util.Scanner;

public class InvoiceDigitProcessing {

    // Method to convert digit to word
    public static String digitToWord(int digit) {
        switch (digit) {
            case 0: return "Zero";
            case 1: return "One";
            case 2: return "Two";
            case 3: return "Three";
            case 4: return "Four";
            case 5: return "Five";
            case 6: return "Six";
            case 7: return "Seven";
            case 8: return "Eight";
            case 9: return "Nine";
        }
    }
}

```

```

        default: return "";
    }
}

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    System.out.print("Enter Invoice Number: ");
    int number = sc.nextInt();

    int temp = number;
    int sum = 0;

    // Calculate sum of digits
    while (temp > 0) {
        sum += temp % 10;
        temp /= 10;
    }

    System.out.println("Sum of digits: " + sum);

    // Print digits of sum in words
    System.out.print("Digits in words: ");
    String sumStr = Integer.toString(sum);

    for (int i = 0; i < sumStr.length(); i++) {
        int digit = sumStr.charAt(i) - '0';
        System.out.print(digitToWord(digit) + " ");
    }

    sc.close();
}
}

```

Output:

Enter Invoice Number: 458

Sum of digits: 17

5. Student Result Processing System

Write a Java program to calculate the average percentage of marks obtained by a student in **5 subjects** and display the final result as Pass or Fail based on average marks.

Sample Input:

78 65 82 70 75

Sample Output:

Average Percentage: 74.0

Result: Pass

```
import java.util.Scanner;

public class StudentResultProcessing {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[] marks = new int[5];
        int sum = 0;

        System.out.println("Enter marks for 5 subjects:");

        for (int i = 0; i < 5; i++) {
            marks[i] = sc.nextInt();
            sum += marks[i];
        }

        double average = sum / 5.0;

        System.out.println("Average Percentage: " + average);

        if (average >= 40) {
            System.out.println("Result: Pass");
        } else {
            System.out.println("Result: Fail");
        }

        sc.close();
    }
}
```

Output:

Enter marks for 5 subjects:

78 65 82 70 75

Average Percentage: 74.0

Result: Pass

6. Warehouse Inventory Matrix

Write a Java program to find the **transpose** of a matrix representing daily stock levels of products in a warehouse.

Sample Input:

10 20 30

40 50 60

Sample Output:

10 40

20 50

30 60

```
import java.util.Scanner;

public class WarehouseInventoryMatrix {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of rows: ");
        int rows = sc.nextInt();

        System.out.print("Enter number of columns: ");
        int cols = sc.nextInt();

        int[][] matrix = new int[rows][cols];
        int[][] transpose = new int[cols][rows];

        System.out.println("Enter matrix elements:");

        // Input matrix
        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < cols; j++) {
                matrix[i][j] = sc.nextInt();
            }
        }

        // Compute transpose
        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < cols; j++) {
                transpose[j][i] = matrix[i][j];
            }
        }
    }
}
```



```

    }

    // Display transpose
    System.out.println("Transpose Matrix:");
    for (int i = 0; i < cols; i++) {
        for (int j = 0; j < rows; j++) {
            System.out.print(transpose[i][j] + " ");
        }
        System.out.println();
    }

    sc.close();
}
}

```

Output:

Enter number of rows: 2

Enter number of columns: 3

Enter matrix elements:

10 20 30

40 50 60

Transpose Matrix:

10 40

20 50

30 60

7. Employee Salary Analysis

Write a Java program to find the **second highest salary** from an array containing employee salaries.

Sample Input:

25000 40000 32000 28000

Sample Output:

Second Highest Salary: 32000

```
import java.util.Scanner;
```

```

public class EmployeeSalaryAnalysis {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of employees: ");
        int n = sc.nextInt();

        int[] salaries = new int[n];

        System.out.println("Enter employee salaries:");
        for (int i = 0; i < n; i++) {
            salaries[i] = sc.nextInt();
        }

        int highest = Integer.MIN_VALUE;
        int secondHighest = Integer.MIN_VALUE;

        for (int i = 0; i < n; i++) {
            if (salaries[i] > highest) {
                secondHighest = highest;
                highest = salaries[i];
            } else if (salaries[i] > secondHighest && salaries[i] < highest) {
                secondHighest = salaries[i];
            }
        }

        if (secondHighest == Integer.MIN_VALUE) {
            System.out.println("Second highest salary not found.");
        } else {
            System.out.println("Second Highest Salary: " + secondHighest);
        }

        sc.close();
    }
}

```

Output:

```

Enter number of employees: 4
Enter employee salaries:
25000 40000 32000 28000
Second Highest Salary: 32000

```

8. Temperature Monitoring System

Write a Java program to find the **index position of the highest temperature** recorded in a day from an array of temperature readings.

Sample Input:

32 35 31 38 36

Sample Output:

Highest Temperature: 38

Index Position: 3

```
import java.util.Scanner;

public class TemperatureMonitoringSystem {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of temperature readings: ");
        int n = sc.nextInt();

        int[] temp = new int[n];

        System.out.println("Enter temperature readings:");
        for (int i = 0; i < n; i++) {
            temp[i] = sc.nextInt();
        }

        int highest = temp[0];
        int index = 0;

        for (int i = 1; i < n; i++) {
            if (temp[i] > highest) {
                highest = temp[i];
                index = i;
            }
        }

        System.out.println("Highest Temperature: " + highest);
        System.out.println("Index Position: " + index);

        sc.close();
    }
}
```

Output:

Enter number of temperature readings: 5

Enter temperature readings:

32 35 31 38 36

Highest Temperature: 38

Index Position: 3

9. Water Consumption Billing System

Write a Java program to calculate the monthly water bill based on consumption units:

- First 200 units → ₹5 per unit
- 201–400 units → ₹6.5 per unit
- Above 400 units → ₹8 per unit

Sample Input:

350

Sample Output:

Total Water Bill: ₹2125

```
import java.util.Scanner;

public class WaterConsumptionBilling {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter water consumption units: ");
        int units = sc.nextInt();

        double bill = 0;

        if (units <= 200) {
            bill = units * 5;
        }
        else if (units <= 400) {
            bill = (200 * 5) + (units - 200) * 6.5;
        }
        else {
            bill = (200 * 5) + (200 * 6.5) + (units - 400) * 8;
        }

        System.out.println("Total Water Bill: ₹" + bill);
    }
}
```

```
        sc.close();
    }
}
```

Output:

Enter water consumption units: 350

Total Water Bill: ₹1975.0

10. Pattern Printing

Problem:

Write a Java program to print the following pattern using numbers:

```
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

Sample Input:

5

Sample Output:

```
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

```
import java.util.Scanner;
```

```
public class PatternPrinting {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        for (int i = n; i >= 1; i--) {
            for (int j = 1; j <= i; j++) {
                System.out.print(j + " ");
            }
        }
    }
}
```

```
        }  
        System.out.println();  
    }  
  
    sc.close();  
}  
}
```

Output:

Enter a number: 5

1 2 3 4 5

1 2 3 4

1 2 3

1 2

1